GCVE-BCP-03 - Decentralized Publication Standard

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Contents

1 Decentralized Publication Standard

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1.1 Introduction

This document describes the decentralized publication model that allows GNAs to publish their vulnerability information directly, without relying on a centralized system.

It also outlines the access methods GNAs use to distribute published vulnerabilities through various mechanisms.

Clients can rely on this BCP document to obtain vulnerabilities published by a GNA.

1.2 Mechanism

The decentralized model is based on the principle that each GNA has full control over its own publication process. The GCVE directory then provides a way to discover the entry points used to collect vulnerability information from a trusted set of GNAs, allowing users to decide whom to trust and from whom to pull vulnerability information.

1.3 Reference Implementation

A reference implementation is available in the open-source project Vulnerability-Lookup, which supports this BCP. It can be used both for decentralized publication and for collecting vulnerability information from all GNAs listed in the GCVE directory.

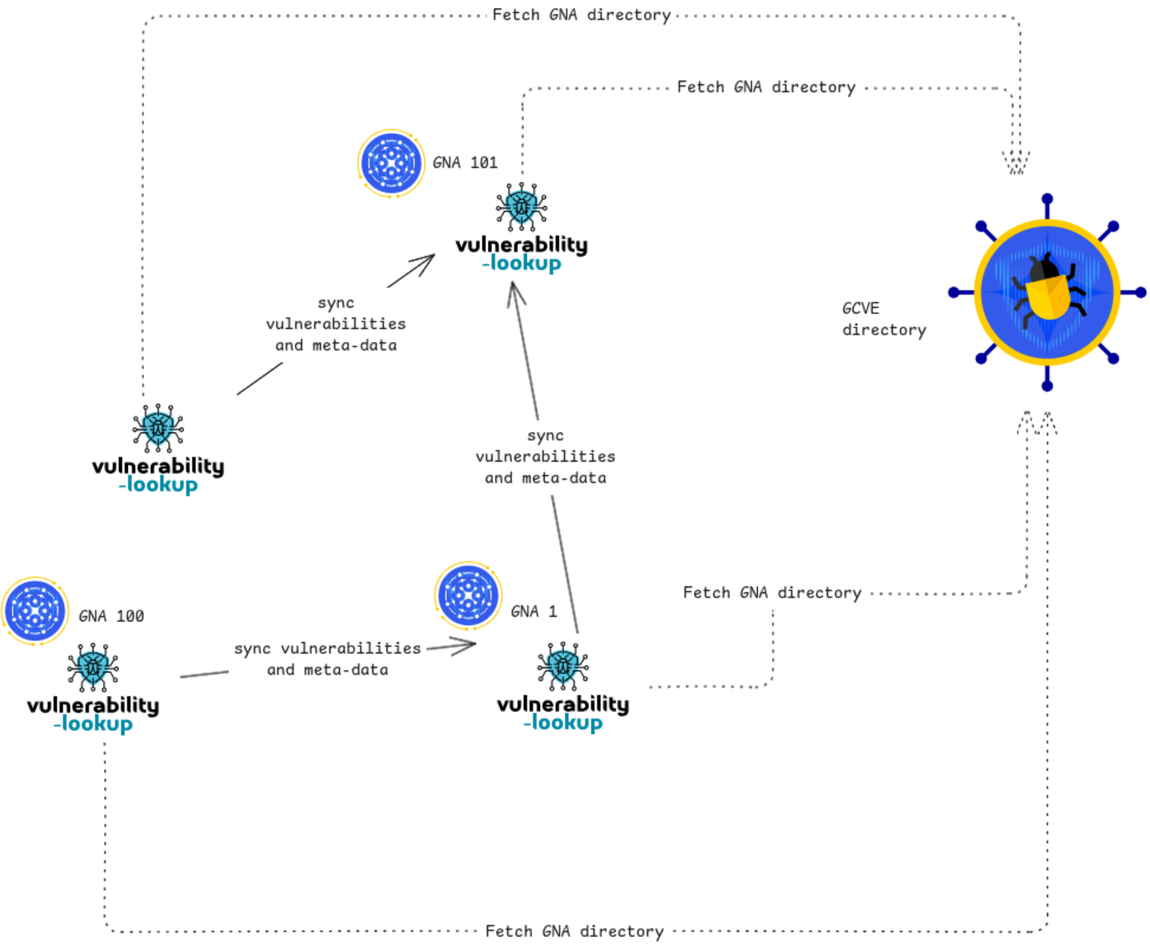

Figure 1.1: An overview of the GCVE decentralized publication model

1.4 Transport

The transport mechanism used to gather vulnerability information relies on HTTP, with two access modes exposed through a single URL. One mode is a simple REST API that allows retrieval of the latest published vulnerabilities, either starting from a specific date or by paginating through the entire dataset. The other mode is a static endpoint that serves a vulnerability file. A GNA can use one or both transport methods.

The URL is referenced in the GCVE directory under the `gcve_pull_api` field.

1.4.1 HTTP REST API

The API base URL is defined in the `gcve_pull_api` field and must support at least the following endpoint:

- `/api/gcve/publication` — Retrieves published vulnerabilities for the local GNA.

The full URL is constructed from the value of the `gcve_pull_api` field in the GCVE directory.

By default, the endpoint **MUST** return only the local GNA's vulnerabilities in the GCVE-BCP-05 format. In addition, a set of optional filters **MUST** be available to allow clients to refine the data returned by the local GNA.

1.4.1.1 Retrieve vulnerabilities with optional filtering and pagination

This endpoint retrieves vulnerabilities with optional filtering and pagination.

1.4.1.1.1 Behavior

- Returns **full vulnerability details** by default.

1.4.1.1.2 Query Parameters

`source` : `str` Optional source used to filter vulnerabilities (for example: `CVE`, `GHSA`, `PySec`).

`per_page` : `int`, default=30 Maximum number of results, capped at `100`.

`date_sort` : `str` Field used for sorting.

Supported values: - `''` - `published` - `updated` - `reserved`

`sort_order` : `str` Sort order.

Supported values: - `asc` - `desc`

`since` : `str` Retrieves vulnerabilities published or updated after the specified date.

`page` : `int` Pagination page number.

cwe : str Filters vulnerabilities by a specific CWE ID.

product : str Optional product name used to filter vulnerabilities (case-insensitive).

If set, the endpoint returns vulnerabilities related to this product across all vendors. Use it with `assigner` to narrow the results further.

assigner : str Optional assigner short name used to filter results (case-insensitive).

This filter is effective only when used with `product` or `vendor`.

1.4.1.1.3 Returns `list[dict[str, Any]] | list[tuple[str, str | None]]`

Returns either: - full vulnerability details, or - minimal tuples if light mode is enabled.

1.4.2 Static File

The full URL of the static endpoint is constructed from the value of the `gcve_pull_api` field in the GCVE directory.

- `/dumps/gna-{GCVE-ID}.ndjson` — A static dump of the vulnerabilities published by the GNA.

A `security.txt` file can be used to declare a GCVE publication endpoint using the `GCVE` field.

1.5 Format

The format must adhere to the CVE Record Format as described in the CVE Record Format schema.

The detailed format is specified in **GCVE-BCP-05**, particularly to ensure compatibility with the CVE Record Format while also supporting the potential extended fields required by GCVE.

1.6 Example Service

`GNA-1` provides a reference service that can be used to query and test a client.

- The API endpoint is available at: https://vulnerability.circl.lu/api/
- The static file is available at: https://vulnerability.circl.lu/dumps/gna-1.ndjson